



ALLIED TELESIS BACKUP CONFIGURATION MANAGEMENT

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Platform : Windows 10/11 (64-bit)
Tanggal : Oktober 2025

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1. PENGENALAN APLIKASI

1.1 Tentang Aplikasi

Allied Telesis Backup Configuration Manager adalah aplikasi desktop untuk melakukan backup konfigurasi switch Allied Telesis secara otomatis dan manual. Aplikasi ini membantu administrator jaringan mengelola, menyimpan, dan membandingkan konfigurasi switch dengan mudah dan aman.

Keunggulan Utama

- ☒ **User-Friendly** : Interface intuitif dengan tema modern
- ☒ **Secure** : Enkripsi kredensial dengan master password
- ☒ **Automated** : Scheduling backup otomatis (harian/mingguan/bulanan)
- ☒ **Real-Time** : Auto-refresh data setiap view
- ☒ **Multi-Protocol** : Support SSH dan Telnet
- ☒ **Comparison** : Diff view untuk bandingkan konfigurasi

2. SPESIFIKASI & INSTALASI

2.1 Kebutuhan Sistem

Komponen	Spesifikasi
OS	Windows 10/11 64-bit
RAM	2-4 GB
Disk	500 MB
Network	Akses ke switch

2.2 Instalasi

1. Extract file AlliedTelesBackup_v1.0.0.zip
2. Jalankan AlliedTelesBackup.exe atau Start.bat
3. Set master password pada first run
4. Aplikasi siap digunakan

Struktur Folder:

text

AlliedTelesBackup/

|— AlliedTelesBackup.exe

|— Start.bat

|— data/ (Database)

|— backups/ (File backup)

|— logs/ (Log aplikasi)

Name	Date modified	Type	Size
backups	10/22/2025 12:33 PM	File folder	
config	10/22/2025 12:33 PM	File folder	
data	10/22/2025 12:49 PM	File folder	
logs	10/22/2025 12:33 PM	File folder	
ATI CM.exe	10/22/2025 12:28 PM	Application	34,196 KB

3. FITUR-FITUR UTAMA

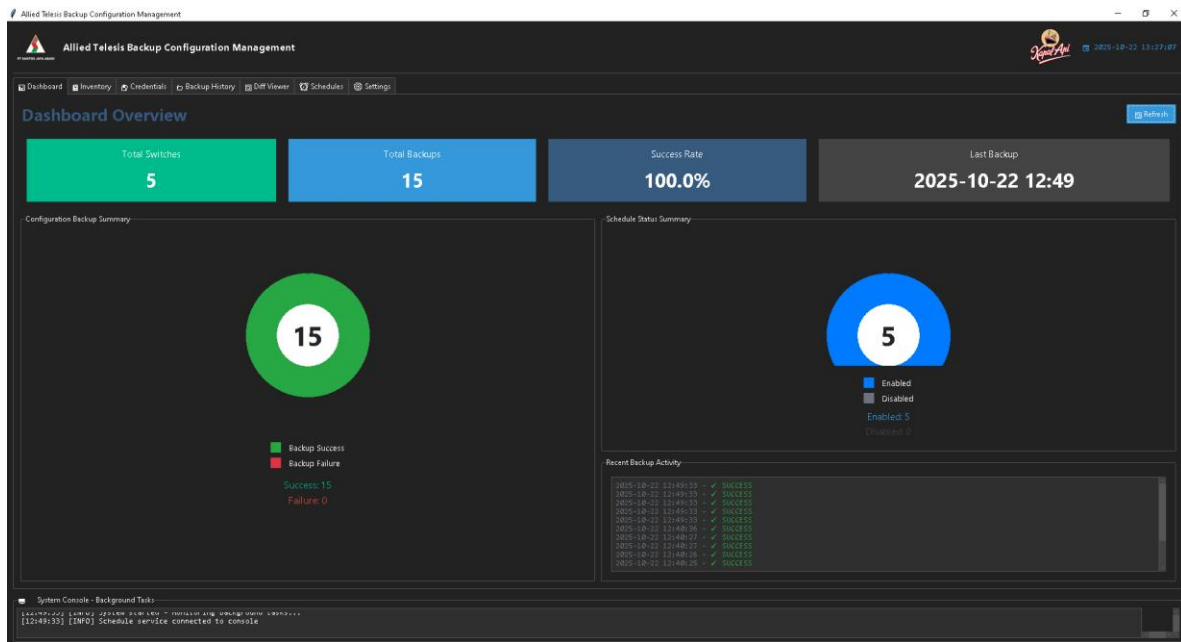
3.1 Dashboard

Fungsi:

- Statistik real-time (Total Switches, Backups, Success Rate)
- Chart visualisasi backup status
- Recent activity log
- Auto-refresh setiap 30 detik

Komponen:

1. **Stat Cards** : 4 kartu statistik utama
2. **Pie Charts** : Backup status & Schedule status
3. **Activity Log** : 10 aktivitas terakhir
4. **Refresh Button**: Manual refresh



3.2 Inventory (Manajemen Switch)

Fungsi:

- Mengelola daftar switch
- Backup manual per switch
- Batch import dari CSV
- Monitoring status backup
- Auto-refresh setiap 30 detik

Fitur Utama:

A. Tabel Switch

Kolom: ID, Name, IP, Protocol, Port, Credential, Last Backup, Status

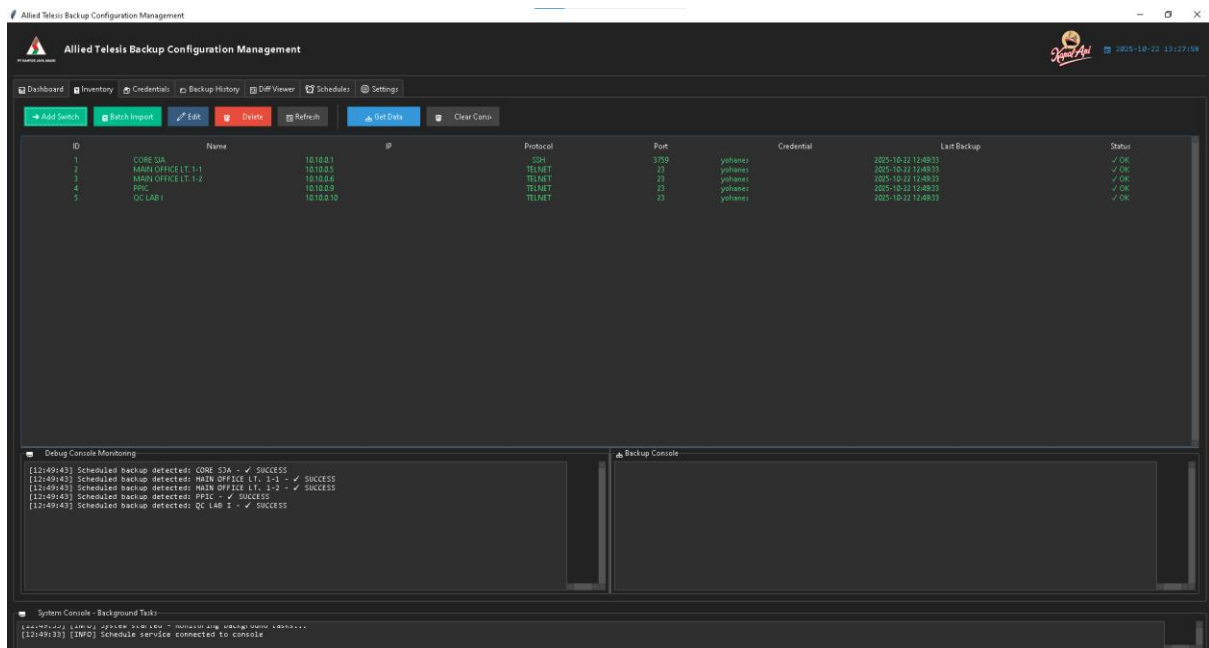
B. Tombol Aksi

-  Add Switch
-  Edit
-  Delete
-  Batch Import
-  Get Data (manual backup)
-  Refresh

C. Dual Console

- Kanan: Output backup manual
- Kiri: Background tasks & system messages

Inventory View



The screenshot displays the 'Allied Telesis Backup Configuration Management' web interface. The top navigation bar includes links for Dashboard, Inventory, Credentials, Backup History, Diff Viewer, Schedules, and Settings. Below the navigation bar, there are several action buttons: Add Switch, Batch Import, Edit, Delete, Refresh, Get Data, and Clear Console. The main content area shows a table with the following data:

ID	Name	IP	Protocol	Port	Credential	Last Backup	Status
1	CORE SW	10.10.0.1	SSH	3759	yohanes	2025-10-22 12:48:33	✓ OK
2	MAIN OFFICE LT-1-1	10.10.0.3	TelNET	23	yohanes	2025-10-22 12:48:33	✓ OK
3	MAIN OFFICE LT-1-2	10.10.0.4	TelNET	23	yohanes	2025-10-22 12:48:33	✓ OK
4	PPIC	10.10.0.9	TelNET	23	yohanes	2025-10-22 12:48:33	✓ OK
5	QC LAB 1	10.10.0.10	TelNET	23	yohanes	2025-10-22 12:48:33	✓ OK

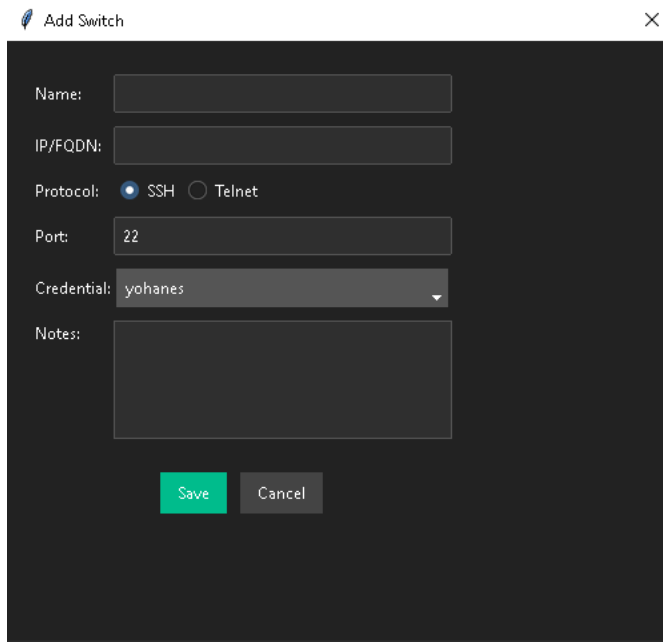
Below the table, there are two console windows. The left window, titled 'Debug Console Monitoring', shows the following log entries:

```
[12:49:43] Scheduled backup detected: CORE SW - ✓ SUCCESS  
[12:49:43] Scheduled backup detected: MAIN OFFICE LT-1-1 - ✓ SUCCESS  
[12:49:43] Scheduled backup detected: MAIN OFFICE LT-1-2 - ✓ SUCCESS  
[12:49:43] Scheduled backup detected: PPIC - ✓ SUCCESS  
[12:49:43] Scheduled backup detected: QC LAB 1 - ✓ SUCCESS
```

The right window, titled 'Backup Console', is currently empty. At the bottom of the interface, there is a 'System Console - Background Tasks' section showing the following log entry:

```
[12:49:33] [INFO] Schedule service connected to console
```

Add Switch Dialog

A dark-themed dialog box titled "Add Switch" with a close button (X) in the top right corner. It contains several input fields: "Name:" with a text box, "IP/FQDN:" with a text box, "Protocol:" with radio buttons for "SSH" (selected) and "Telnet", "Port:" with a text box containing "22", "Credential:" with a dropdown menu showing "yohanes", and "Notes:" with a large text area. At the bottom are two buttons: "Save" (green) and "Cancel" (grey).

Add Switch

Name:

IP/FQDN:

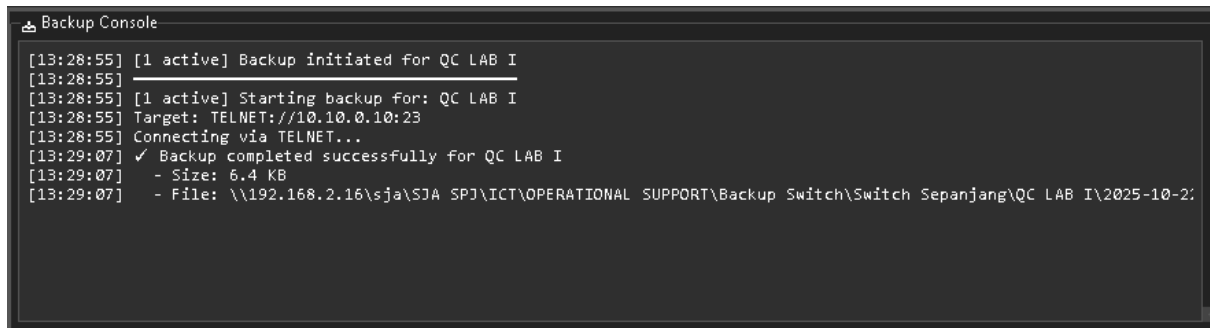
Protocol: ☒ SSH ☐ Telnet

Port:

Credential:

Notes:

Manual Backup Console

A dark-themed console window titled "Backup Console" showing a log of backup operations. The log includes timestamps, status messages, and file paths.

```
[13:28:55] [1 active] Backup initiated for QC LAB I
[13:28:55] [1 active] Starting backup for: QC LAB I
[13:28:55] Target: TELNET://10.10.0.10:23
[13:28:55] Connecting via TELNET...
[13:29:07] ✓ Backup completed successfully for QC LAB I
[13:29:07] - Size: 6.4 KB
[13:29:07] - File: \\192.168.2.16\sja\SJA SPJ\ICT\OPERATIONAL SUPPORT\Backup Switch\Switch Sepanjang\QC LAB I\2025-10-2:
```

3.3 History (Riwayat Backup)

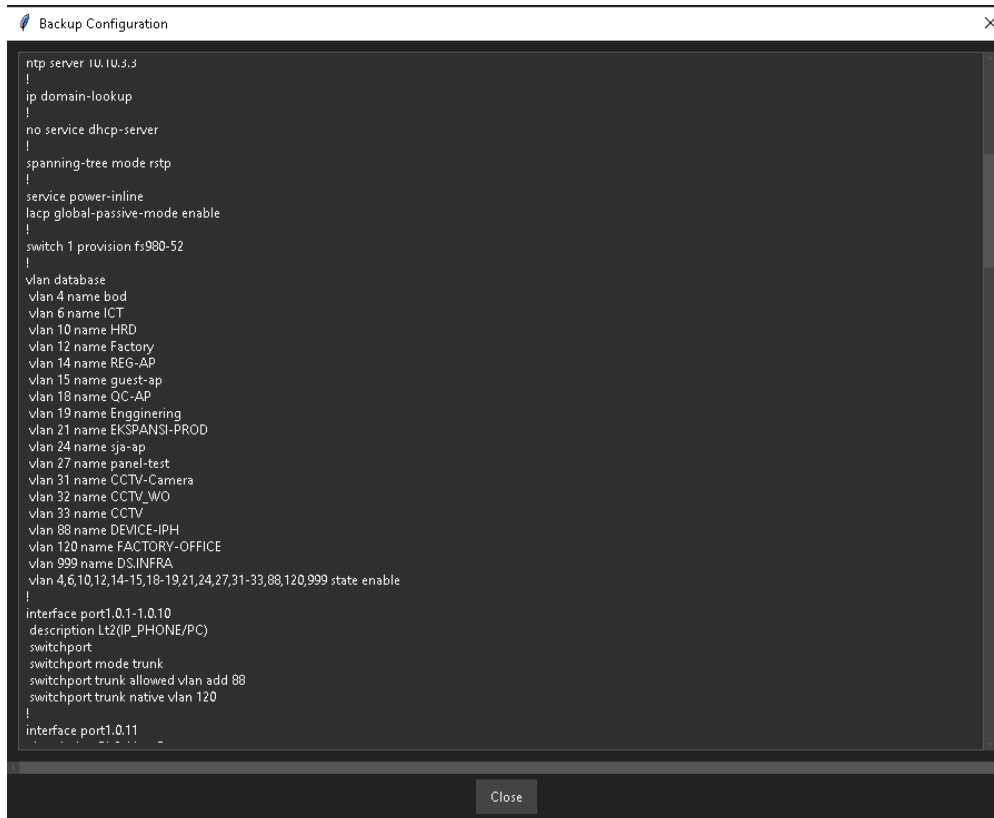
Fungsi:

- Menampilkan history semua backup
- Filter: Switch, Type (Manual/Auto), Month, Search
- View, Export, dan Open folder backup
- Auto-refresh setiap 30 detik

Color Coding:

- **Hijau** : Auto backup sukses
- **Cyan** : Manual backup sukses
- **Merah** : Auto backup gagal
- **Orange**: Manual backup gagal

Backup Content Viewer



The screenshot shows a window titled "Backup Configuration" with a close button (X) in the top right corner. The main area displays a configuration file with the following content:

```
ntp server 10.10.3.3
!
ip domain-lookup
!
no service dhcp-server
!
spanning-tree mode rstp
!
service power-inline
lACP global-passive-mode enable
!
switch 1 provision fs980-52
!
vlan database
vlan 4 name bod
vlan 6 name ICT
vlan 10 name HRD
vlan 12 name Factory
vlan 14 name REG-AP
vlan 15 name guest-ap
vlan 18 name QC-AP
vlan 19 name Engginering
vlan 21 name EKSPANSI-PROD
vlan 24 name sja-ap
vlan 27 name panel-test
vlan 31 name CCTV-Camera
vlan 32 name CCTV_WO
vlan 33 name CCTV
vlan 88 name DEVICE-IPH
vlan 120 name FACTORY-OFFICE
vlan 999 name DS.INFRA
vlan 4,6,10,12,14-15,18-19,21,24,27,31-33,88,120,999 state enable
!
interface port1.0.1-1.0.10
description Lt2(IP_PHONE/PC)
switchport
switchport mode trunk
switchport trunk allowed vlan add 88
switchport trunk native vlan 120
!
interface port1.0.11
```

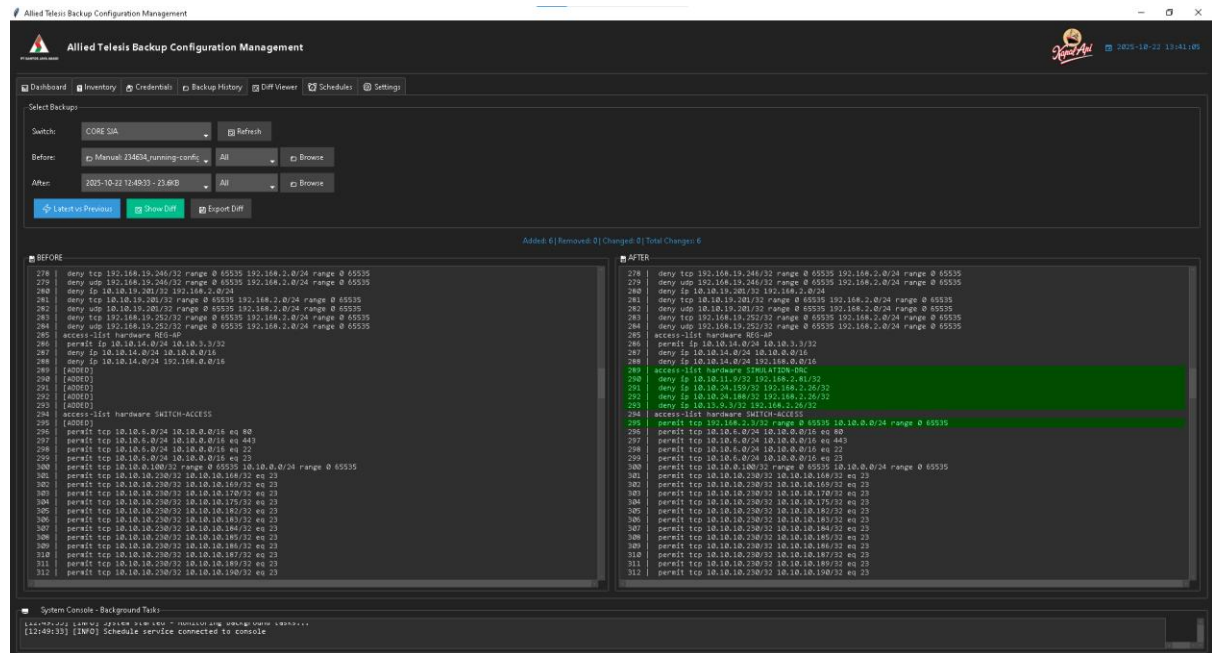
At the bottom of the window, there is a "Close" button.

3.4 Diff View (Perbandingan Konfigurasi)

Fungsi:

- Membandingkan 2 konfigurasi switch
- Filter month individual untuk Before & After
- Manual file browse (compare file dari komputer)
- Side-by-side comparison dengan color coding
- Export hasil diff

Diff View Selection



3.5 Schedules (Penjadwalan)

Fungsi:

- Membuat jadwal backup otomatis
- Interval: Minutes, Hours, Daily, Weekly, Monthly
- Enable/Disable schedule
- Start Now (manual trigger)
- Auto-refresh setiap 10 detik

Interval Types:

- **Minutely:** Every X minutes (1-59)
- **Hourly:** Every X hours (1-23)
- **Daily:** Every day at HH:MM
- **Weekly:** Every 7 days at HH:MM
- **Monthly:** Every 30 days at HH:MM

Schedules View

Dashboard Inventory Credentials Backup History Diff Viewer Schedules Settings

+ Add Schedule Edit Delete Enable Disable Start Now Refresh

Statistics

Total Schedules: 5 | Active: 5 | Successful Backups (Last 30 days): 17

ID	Switch	Interval	Enabled	Last Run	Next Run
1	CORE SJA	Every 12 hours	✓	2025-10-22 12:49:33	2025-10-23 00:49:12
2	MAIN OFFICLT 1-1	Every 1 day	✓	2025-10-22 12:49:33	2025-10-23 06:00:00
3	MAIN OFFICLT LT 1-2	Every 1 day	✓	2025-10-22 12:49:33	2025-10-23 06:00:00
4	PPIC	Every 1 day	✓	2025-10-22 12:49:33	2025-10-23 06:00:00
5	QC LAB 1	Every 1 day	✓	2025-10-22 12:49:33	2025-10-23 06:00:00

System Console - Background Tasks

```
12/22/2025 12:49:33 [INFO] Backup service - Initializing backup service...
12/22/2025 12:49:33 [INFO] Schedule service connected to console
```

Add Schedule Dialog

Add Schedule

Switch: CORE SJA

Schedule Type: ☒ Interval ☐ Daily ☐ Weekly ☐ Monthly

Interval Options

☐ Every 15 minutes

☒ Every hour

☐ Every 6 hours

☐ Every 12 hours

Save Cancel

3.6 Credentials (Manajemen Kredensial)

Fungsi:

- Mengelola username/password switch
- Enkripsi dengan Fernet (symmetric encryption)
- Add, Edit, Delete credentials
- Enable password optional (privilege escalation)
- Auto-refresh setiap 60 detik

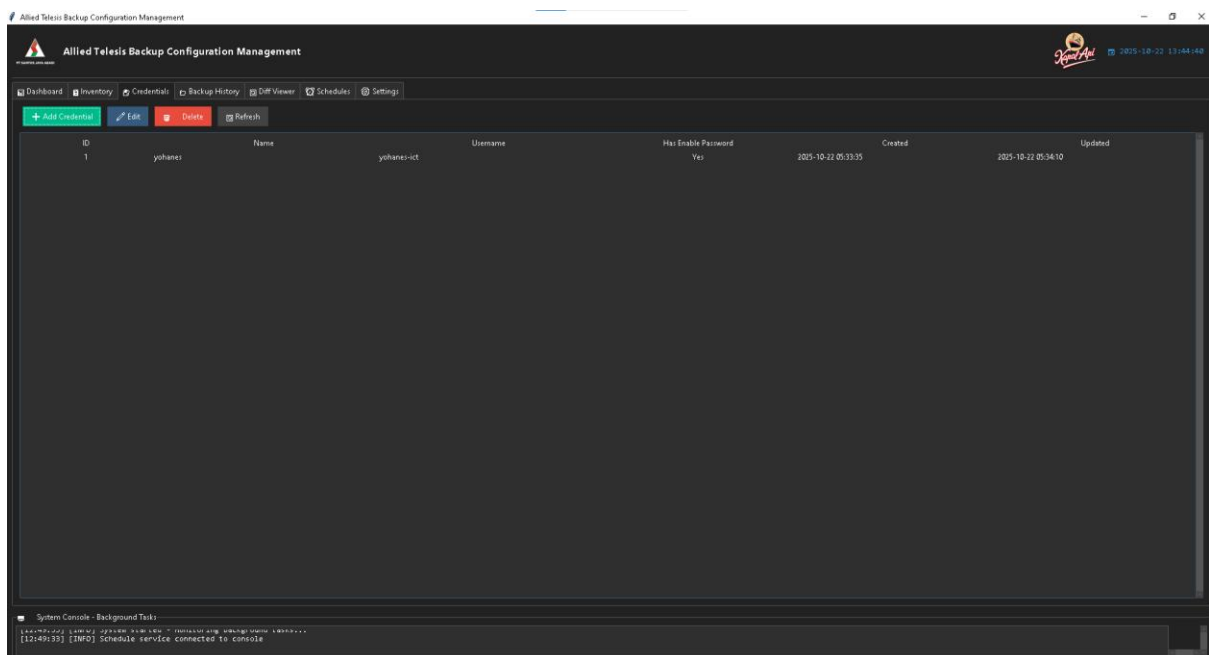
Tabel Credentials:

Kolom: ID, Name, Username, Has Enable Password, Created, Updated

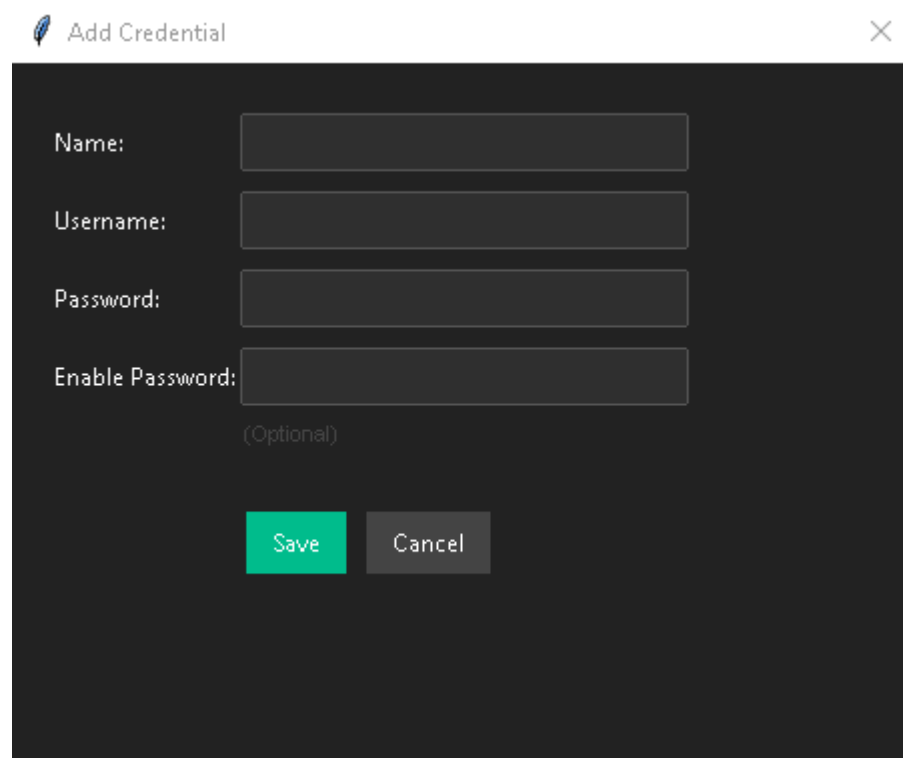
Security:

- Password dienkripsi
- Master password protection
- No plain text storage

Credentials View



Add Credential Dialog



The image shows a dark-themed dialog box titled "Add Credential" with a feather icon and a close button (X) in the top right corner. The dialog contains four input fields: "Name:", "Username:", "Password:", and "Enable Password:". The "Enable Password:" field has a small "(Optional)" label below it. At the bottom, there are two buttons: a green "Save" button and a grey "Cancel" button.

Add Credential

Name:

Username:

Password:

Enable Password:

(Optional)

3.7 Settings (Pengaturan)

A. Backup Settings

- Retention days (default: 365)
- Backup path
- Compression

B. Network Settings

- Connection timeout
- Retry attempts
- Command timeout

C. Security

- Change master password

E. Logging

- Log level

- Log retention

Settings View

**Allied Telesis Backup Configuration Management**

[Dashboard](#) [Inventory](#) [Credentials](#) [Backup History](#) [Diff Viewer](#) [Schedules](#) [Settings](#)

Application Settings

Backup Retention

Backups are automatically cleaned up after 365 days (1 year).
At least 1 backup per switch is always retained.

 [Run Cleanup Now](#)

Network Settings

Connection Timeout: 15 seconds
Read Timeout: 30 seconds
Command Timeout: 60 seconds
Max Retries: 3

[Edit app/config/appsettings.yaml](#) to customize.

CLI Prompt Patterns

Recognized CLI prompts for ATi switches:

```
# - Privileged exec mode
> - User exec mode
(config)# - Global configuration mode
(config-if)# - Interface configuration mode
```

[Edit app/config/appsettings.yaml](#) to customize prompt patterns.

Backup Storage Location

Configure where backup configurations are saved:

Current Path:

[Browse](#)



Allied Telesis Backup Configuration Management

[Dashboard](#) [Inventory](#) [Credentials](#) [Backup History](#) [Diff Viewer](#) [Schedules](#) [Settings](#)

Logo Configuration

Configure company and application logos for branding:

Company Logo: [Browse](#) [Clear](#)

Application Logo: [Browse](#) [Clear](#)

Supported formats: PNG, JPG, GIF (Recommended: 200x60 pixels for best display)

Security

Change your master passphrase used to encrypt credentials.
Warning: All existing encrypted credentials will be re-encrypted.

[Change Master Password](#) [Factory Reset](#)

System - Time Synchronization

NTP Server: (e.g., pool.ntp.org, time.google.com)

Current System Time: 2025-10-22 12:49:33

[Sync Time with NTP Serve](#)

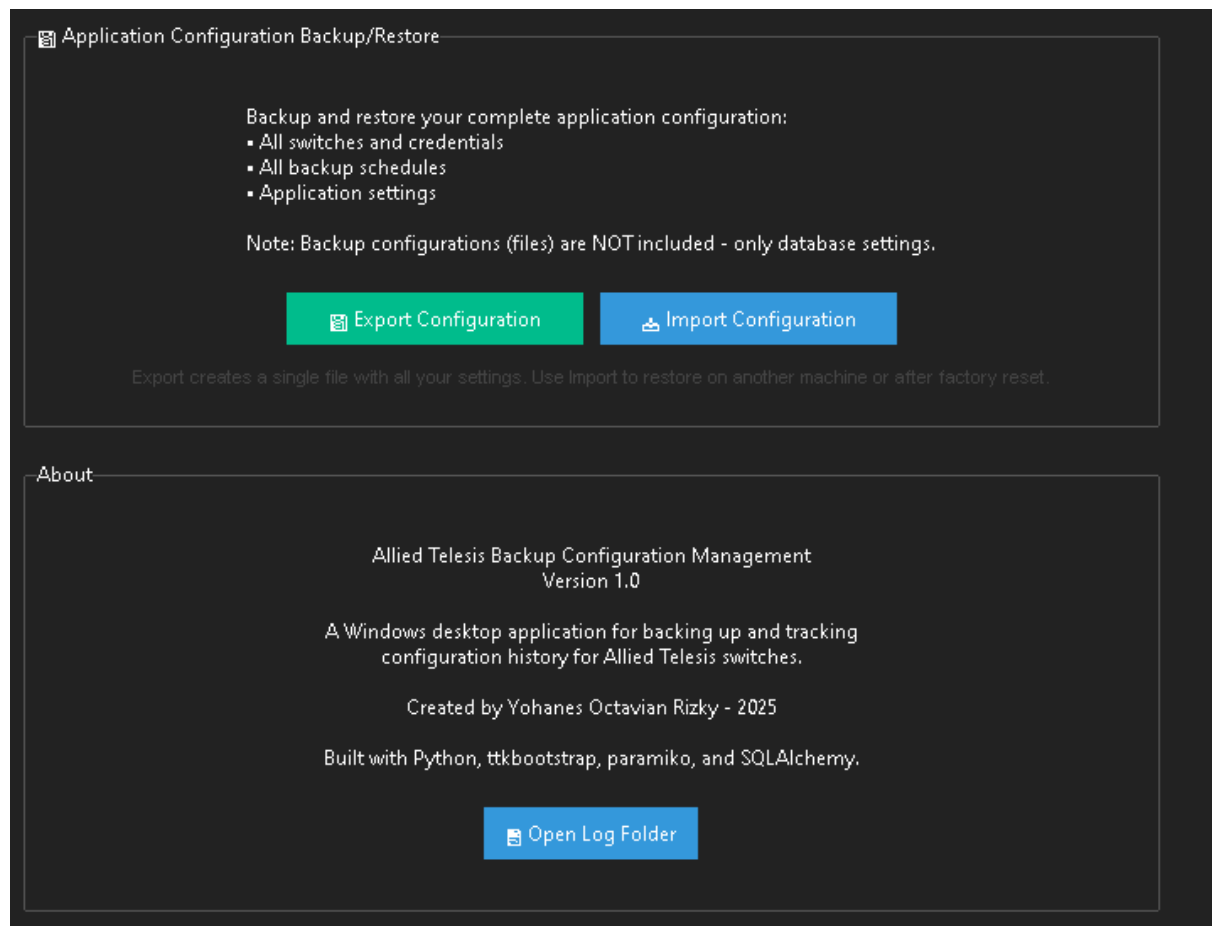
Note: Requires administrator privileges on Windows to change system time

Auto-Start Service Control

Configure automatic startup using Windows Task Scheduler (reliable, no errors!):

Task Status: ✔ Configured [Refresh Status](#)

[Setup Auto-Start](#) [Start Service Now](#) [Remove Auto-St:](#)



4. PANDUAN PENGGUNAAN

4.1 Quick Start (5 Menit)

LANGKAH 1: Setup Awal

1. Jalankan aplikasi
2. Set master password (simpan baik-baik!)
3. Konfirmasi password

LANGKAH 2: Tambah Credential

1. Buka tab **Credentials**
2. Klik **+ Add Credential**

3. Isi:

- Name: admin_ssh
- Username: manager
- Password: [password switch Anda]

4. **Save**

LANGKAH 3: Tambah Switch

1. Buka tab **Inventory**

2. Klik **+** **Add Switch**

3. Isi:

- Name: Switch-Core-01
- IP: 192.168.1.1
- Protocol: SSH
- Port: 22
- Credential: admin_ssh

4. **Save**

LANGKAH 4: Test Backup

1. Pilih switch dari tabel
2. Klik  **Get Data**
3. Monitor console untuk progress
4. Cek tab **History** untuk hasil

LANGKAH 5: Setup Schedule

1. Buka tab **Schedules**
2. Klik **+** **Add Schedule**
3. Pilih switch: Switch-Core-01
4. Interval: Every 1 day at 02:00

5. Centang **Enabled**

6. **Save**

4.2 Workflow Backup Manual

text

1. Inventory → Pilih Switch → Get Data
2. Monitor Console untuk progress
3. Cek status di kolom "Last Backup"
4. Lihat hasil di tab History
5. (Optional) View atau Export backup

4.3 Workflow Backup Otomatis

text

1. Credentials → Add credential
2. Inventory → Add switch
3. Schedules → Add schedule → Enable
4. System akan backup otomatis sesuai schedule
5. Monitor di Dashboard (Recent Activity)
6. Check hasil di History

4.4 Workflow Compare Konfigurasi

Scenario 1: Compare Database Backups

text

1. Diff View → Pilih Switch
2. Before: Pilih backup lama
3. After: Pilih backup baru
4. Show Diff
5. Analisa perbedaan (red/green/yellow)

6. (Optional) Export Diff

Scenario 2: Compare dengan File Manual

text

1. Diff View → Pilih Switch
2. Before: Pilih dari database
3. After: Click  Browse → Pilih file
4. Show Diff
5. Analisa perbedaan

5. KEBUTUHAN BACKUP

5.1 Mengapa Perlu Backup Konfigurasi?

Alasan Utama:

1. Disaster Recovery

- Switch failure → Restore konfigurasi cepat
- Hardware replacement → Apply config ke switch baru
- Minimize downtime

2. Audit & Compliance

- Track perubahan konfigurasi
- Compliance reporting

3. Change Management

- Bandingkan before/after perubahan
- Rollback jika ada masalah
- Documentation perubahan

4. Security

- Detect unauthorized changes
- Backup sebelum maintenance

- Investigation security incidents

5. Best Practice

- Industry standard
 - Prevent data loss
 - Peace of mind
-

5.2 Strategi Backup

Rekomendasi Interval:

Production Switches (Critical):

- **Daily:** Backup setiap 12 jam
- **Before Change:** Manual backup sebelum perubahan
- **After Change:** Manual backup setelah perubahan
- **Retention:** 90 hari

Distribution Switches:

- **Weekly:** Backup setiap hari
- **Before Change:** Manual backup sebelum perubahan
- **Retention:** 60 hari

Access Switches:

- **Monthly:** Backup setiap hari
- **Ad-hoc:** Manual backup saat perubahan
- **Retention:** 30 hari

LAMPIRAN

Format CSV untuk Batch Import

CSV

name,ip,protocol,port,credential_name,notes
Switch-Core-01,192.168.1.1,ssh,22,admin_ssh,Main Core Switch
Switch-Dist-01,192.168.1.10,ssh,22,admin_ssh,Distribution Floor 1
Switch-Access-01,192.168.1.100,telnet,23,user_telnet,Access Switch Lab

Command Reference

Command	Function
Ctrl + R	Refresh current view
Ctrl + F	Search/Filter
Ctrl + S	Save
Escape	Close dialog
F1	Help
F5	Manual refresh

File Locations

Item	Location
Database	data/backups.db
Backups	backups/[switch_name]/
Logs	logs/app_[date].log
Config	app/config/appsettings.yaml

Dokumentasi ini dibuat untuk aplikasi Allied Telesis Backup Configuration Manager v1.0.0

Terakhir diupdate: Oktober 2025

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